

Solution

FORCE-04

Class 09 - Science

Section A

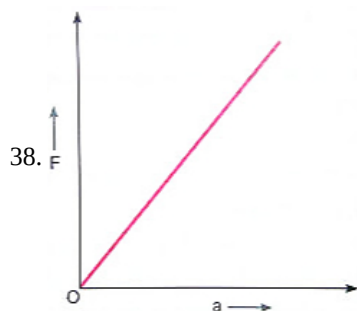
1. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: While catching a fast-moving cricket ball, a fielder in the ground gradually pulls his hands backwards. While doing so, the fielder increases the time during which the high velocity of the moving ball decreases to zero. Thus, the acceleration of the ball is decreased and therefore the impact of catching the fast-moving ball is also reduced.
2. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Newton's first law of motion states that a body at rest will remain at rest, and a body in motion will continue in motion in a straight line with a uniform speed unless it is compelled by an external force to change its state of rest or of uniform motion.
All objects resist a change in their state of motion. The tendency of undisturbed objects to stay at rest or to keep moving with the same velocity is called inertia. This is why the first law of motion is also known as the law of inertia.
3. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
4. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
5. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: According to Newton's second law of motion, $a = \frac{F}{m}$ i.e., the magnitude of the acceleration produced by a given force is inversely proportional to the mass of the body. Higher is the mass of the body, lesser will be the acceleration produced i.e., the mass of the body is a measure of the opposition offered by the body to change a state when the force is applied i.e., the mass of a body is the measure of its inertia.
6. **(c)** A is true but R is false.
Explanation: A is true but R is false.
7. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
8. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The momentum (p) of a body is defined as the product of its mass (m) and velocity (v) and has the same direction as that of the velocity.
Momentum = mass $\times$ velocity
 $p = m \times v$
Hence, its SI unit is kilogram metre per second (kg m s^{-1}).
9. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: Both A and R are true but R is not the correct explanation of A.
10. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: The third law of motion states that when one object exerts a force on another object, the second object instantaneously exerts a force back on the first. These two forces are always equal in magnitude but opposite in direction. The two opposing forces are also known as action and reaction forces.
11. **(c)** A is true but R is false.
Explanation: Inertia is that property of the body due to which it resists a change in its state of rest or of uniform motion. Heavy objects have more inertia than lighter objects. For example, if we kick a stone it will not move because of high inertia but if we kick a football, it will move a long way.

12. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Both A and R are true and R is the correct explanation of A.
13. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: A force can change the direction of a moving body. For example - in a cricket match, when a cricket ball is hit by a bat, then the direction of the cricket ball changes and it goes in another direction.
14. **(c)** A is true but R is false.
Explanation: According to third law of motion it is impossible to have a single force out of mutual interaction between two bodies, whether they are moving or at rest. While Newton's third law is applicable for all types of forces.
15. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: When a gun is fired, it exerts a forward force on the bullet. The bullet exerts an equal and opposite reaction force on the gun. This results in the recoil of the gun.
16. **(c)** A is true but R is false.
Explanation: $\text{acceleration} = \frac{\text{force}}{\text{mass}}$ i.e.,
 if net external force on the body is zero ($F = 0$), then, the acceleration of a body is also zero.
17. **(b)** Both A and R are true but R is not the correct explanation of A.
Explanation: If the resultant of all the forces acting on a body is zero, then the forces are called balanced forces. Suppose a heavy box is lying on the ground. Let us push this box with our hands but it does not move. Though the box is at rest, four forces are acting on it – the force of the push, the force of friction, the force of gravity, and force of reaction. The forces acting on a stationary box is an example of balanced forces. Thus balanced forces cannot produce motion in a stationary body or stop a moving body.
18. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: Heavy objects have more inertia than lighter objects. For example, if we kick a stone it will not move because of high inertia but if we kick a football, it will move a long way. Thus the inertia of an object is measured by its mass.
19. **(c)** A is true but R is false.
Explanation: A is true but R is false.

Section B

20. According to Newton's second law of motion, the rate of change of momentum of a body is equal to the force acting on it and the change in momentum takes place in the same direction as the force applied.
21. The action and reaction act simultaneously but on different bodies. Therefore, the body moves under the action of the acting force.
22. We define one newton as that force which when acting on a mass of 1 kg produces in it an acceleration of 1 m/s^2 in its own direction.
 $1\text{ N} = 1\text{ kg m/s}^2$
23. Yes, whenever the bodies are in actual contact or even if there is an interaction between the bodies (e.g., attraction or repulsion between two magnets, charges, etc.), Newton's third law is applicable.
24. Balanced forces usually change the shape of the body but not the direction of the moving body. They do not change the state of motion of the body, nor they produce any acceleration.
25. Dyne
26. The force of attraction or repulsion of particles or objects because of their electric charge is known as electrostatic force. It is also known as Coulomb force or Coulomb interaction.
27. This is because of the frictional forces acting opposite to the direction of motion. It is the result of the retarding action of friction.
28. The force acting between any two surfaces in contact and tending to oppose motion is called the force of friction. These forces are mainly affected by the surface texture and amount of force impelling them together.
29. It is easier to pull a lighter body than a heavier body. Therefore, more the mass more will be the inertia. That is why mass is sometimes termed as coefficient of linear inertia. Inertia is directly proportional to the mass of the body.
30. Newton's second law of motion gives the measure of force.
31. It is because, the mug experiences a buoyant force equal the weight of water displaced by it. Thus, it feels lighter.
32. Galileo Galilei. (Galileo's law of inertia)

33. Newton's first law of motion states that, 'Every body continues to be in its state of rest or of uniform motion in a straight line unless it is compelled by some external applied force to change that state.'
34. Body A has more inertia. It is because inertia is directly proportional to the mass of a body. More precisely, we say that mass is the measure of the inertia of a body.
35. Yes, the body will gradually slow down due to the opposing force of friction.
36. Law of conservation of momentum is applicable to isolated system (no external force is applied). In this case, the change in velocity is due to the gravitational force of earth.
37. No, action and reaction forces act simultaneously but on different bodies. Action and reaction forces occur in pairs only.



39. No, If two equal and opposite forces act on an object, they both cancel out each other. Hence, there will be no motion. The object will remain in the state of rest.
40. A body continues to move with the same velocity if no unbalanced force acts on it. It stays at rest or moves in a straight line with constant speed, i.e. it will remain unaccelerated.
41. Unbalanced force. Any body moving at constant speed gets accelerated or de-accelerated due to the action of unbalanced forces.
42. When two or more forces act on a body simultaneously, then the single force which produces the same effect as produced by all the forces acting together is known as the resultant force.
43. A force of 200 N is applied in the forward direction. Thus, from Newton's third law of motion, an equal amount of force will act in the opposite direction. Hence, a frictional force of 200 N is exerted on the cabinet.

44. i. Given mass, $m = 50 \text{ g} = \frac{50}{1000} \text{ kg}$.

$$\text{Acceleration during intervals 0 to 3 s} = a_1 = \frac{v-u}{t} = \frac{120-0}{3} = 40 \text{ m/s}^2$$

$$\text{According to Newton's second law of motion : Force, } F_1 = ma = \left(\frac{50}{1000}\right) \times 40 = 2 \text{ N}$$

ii. Acceleration during intervals 6 to 10 s $= a_2 = \frac{v_2-v_1}{t} = \frac{0-120}{(10-6)} = -\frac{120}{4} = -30 \text{ m/s}^2$

$$\text{Similarly, Force, } F_2 = ma_2 = \frac{50}{1000} \times (-30) = -1.5 \text{ N.}$$

- iii. Time interval in which no force acts on the object = '3's - '6' s i.e A to B.

This is because in this interval, the velocity of object is constant i.e. 120 m/s .

Hence, Acceleration = '0' m/s^2 . Therefore, $F = '0' \text{ N}$.

45. i. Acceleration = Slope of the line of the velocity-time graph,

$$a = \frac{v_2-v_1}{t-t_1} = \frac{5-0}{2-0} = \frac{5}{2} = \frac{10}{4} = \frac{15}{6} = 2.5 \text{ m/s}^2$$

- ii. The force acting on the body is given by

$$F = ma = 5 \times 2.5 = 12.5 \text{ N}$$

- iii. $\therefore$ Change in momentum = $mv - mu$ [$\because u = 0$ and $v = 5 \text{ m/s}$]

$$= 5 \times 5 - 5 \times 0$$

$$= 25 \text{ kg-m/s}$$

46. Initial velocity, $u = 360 \text{ m/s}$

Final Velocity, $v = 0$

Distance Travelled, $S = 10 \text{ cm} = 0.1 \text{ m}$

Acceleration, $a = ?$

From third equation of motion, $v^2 = u^2 + 2as$

$$\Rightarrow 0 = (360)^2 + 2a(0.1)$$

$$\Rightarrow a = \frac{-129600}{2} = -648000 \text{ m/s}^2$$

Now,

mass of the bullet, $m = 2.0 \text{ g} = 0.002 \text{ kg}$

Force, $F = ma$

$$= 0.002 \times (-648000)$$

$$= -1296 \text{ N}$$

∴ The average de-accelerating force exerted by the wood = -1296 N

(b)

From first equation of motion, $v = u + at$

$$\Rightarrow 0 = 360 + (-648000)t$$

$$\Rightarrow t = 5.56 \times 10^{-4} \text{ s}$$

Therefore, time taken by the bullet to come to rest = $5.56 \times 10^{-4} \text{ s}$

47. i. Road accidents occurring due to high speeds are much worse than accidents due to low speeds of vehicles. This is because the momentum of high-speed vehicles is more than that of the low speeds of vehicles.
 ii. When a motorcar makes a sharp turn left or right at a high-speed. The lower portion of their passenger turns suddenly along with the motorcar but your upper portion does not change its direction due to inertia.
 So, this portion of a passenger moves forward and the passenger tends to get thrown to one side or another side.

48. The velocity-time graph shows that the velocity of the ball at $t = 0$, is zero. Initial velocity of ball, $u = 0$

Velocity of ball at $t = 4 \text{ s}$ is 20 ms^{-1}

That is, final velocity, $v = 20 \text{ ms}^{-1}$

Time, $t = 4 \text{ s}$.

Acceleration of the ball, $a = \frac{v-u}{t}$

$$\Rightarrow a = \frac{20 \text{ ms}^{-1} - 0 \text{ ms}^{-1}}{4 \text{ s}} \Rightarrow a = 5 \text{ ms}^{-2}$$

Also, mass of ball,

$$m = 100 \text{ g} = \frac{100}{1000} \text{ kg} = \frac{1}{10} \text{ kg}$$

∴ Force acting on the ball, $F = ma$

$$\Rightarrow F = \frac{1}{10} \text{ kg} \times 5 \text{ ms}^{-2}$$

$$= 0.5 \text{ kgms}^{-2} = 0.5 \text{ N} [1 \text{ kgms}^{-2} = 1 \text{ N}]$$

Therefore, the force acting on the ball is 0.5 N .

49. i. While walking on sand, the sand gets pressed down, impacting less reaction force on the person.
 ii. The mass of the heavy gun is higher. So, its recoil velocity is less.

50. i. Distance travelled by car in first 2 s = Area of $\triangle OAD = \frac{1}{2} \times 2 \times 15 = 15 \text{ m}$

ii. Braking force, $F = m \times a$

Given, mass of the car, $m = 1000 \text{ kg}$, initial velocity, $u = 15 \text{ m/s}$, final velocity, $v = 0$, time, $t = 1 \text{ s}$

$$\text{On applying, } a = \frac{v-u}{t} \Rightarrow a = \frac{0-15}{1} = -15 \text{ m/s}^2$$

$$\therefore F = m a = 1000 \times (-15) = -15000 \text{ N}$$

51. Kiran's suggestion is wrong because the mass of insect is very small and velocity of insect will be same as that of motor car after collision; hence change in momentum of insect cannot be greater than that of a motor car. Akhtar's statement is also wrong; because the force is not due to larger velocity, it is due to change in momentum.

52. The velocity-time graph shows that velocity of the ball at $t = 0$ is 30 ms^{-1}

Initial velocity of the ball, $u = 30 \text{ ms}^{-1}$

The velocity of the ball at $t = 6 \text{ s}$ is zero.

Final velocity of the ball, $v = 0$

Time, $t = 6 \text{ s}$

∴ Acceleration of the ball,

$$a = \frac{v-u}{t} = \frac{0-30 \text{ ms}^{-1}}{6 \text{ s}}$$

$$= -5 \text{ ms}^{-2}$$

Negative sign shows that the ball is retarded or decelerated.

$$\text{Also, mass of ball, } M = 50 \text{ g} = \frac{50}{1000} = \frac{1}{20} \text{ kg}$$

Therefore, Force acting on the ball, $F = ma$

$$= \left(\frac{1}{20} \text{ kg} \right) (-5 \text{ ms}^{-2})$$

$$= -0.25 \text{ kg ms}^{-2}$$

$$= 0.25 \text{ N} [1 \text{ kg ms}^{-2} = 1 \text{ N}]$$

Here -ve sign indicates that the force is retarding or stopping force.

53. i. Here, initial velocity, $u = 0$

Using Newton's second law of motion, $s = ut + \frac{1}{2}at^2 = \frac{1}{2}at^2$ [$\because u = 0$]

We get, $a = \frac{2s}{t^2}$

Time (in second)	Distance (in metre)	$a = 2s/t^2$
0	0	0
1	1	2
2	8	4
3	27	6
4	64	8
5	125	10
6	216	12
7	343	14

Thus, acceleration is increasing.

- ii. Since acceleration is increasing, so the net unbalanced force is acting on the object.

54.

Time t (s)	Distance s (m)	Velocity $u = \frac{s}{t}$ (ms^{-1})	Acceleration $a = \frac{v-u}{t}$ (ms^{-2})
0	0	0	-
1	1	1	$a_1 = \frac{1-0}{1-0} = 1$
2	8	4	$a_2 = \frac{4-1}{2-1} = 3$
3	27	9	$a_3 = \frac{9-4}{3-2} = 5$
4	64	16	$a_4 = \frac{16-9}{4-3} = 7$
5	125	25	$a_5 = \frac{25-16}{5-4} = 9$
6	216	36	$a_6 = \frac{36-25}{6-5} = 11$
7	343	49	$a_7 = \frac{49-36}{7-6} = 13$

- i. There is an unequal change of distance in equal interval of time. Thus, the given object is having a non-uniform motion. From the above table, it is clear that the acceleration is not constant and increase with time.

- ii. The object is in accelerated condition. According to Newton's second law of motion, the force acting on an object is directly proportional to the acceleration produced in the object. So, we can say that an unbalanced force is acting on the object.

55. i. 15 g-wt.

- ii. From Newton's third law, the force exerted by B on A and force exerted by A on B are equal.

- iii. Force of reaction balance A exerts on balance B and force of action balance B exerts on balance A.

56. The velocity-time graph shows that the velocity of the ball at $t = 0$ is zero. So, the initial velocity of the ball, $u = 0$.

Velocity of the ball at $t = 4$ s is 20 m/s

i.e. final velocity, $v = 20$ m/s; time, $t = 4$ s

$\therefore$ Acceleration of the ball,

$$a = \frac{v-u}{t} = \frac{20\text{m/s}-0}{4\text{s}} = 5\text{m/s}^2$$

Also, mass of the ball

$$a = 100\text{g} \Rightarrow \frac{100}{1000}\text{kg} = \frac{1}{10}\text{kg}$$

$\therefore$ Force acting on the ball,

$$F = ma$$

$$F = \frac{1}{10}\text{kg} \times 5\text{m/s}^2$$

$$= 0.5\text{ kg-m/s}^2 = 0.5\text{ N} [\because 1\text{ kg-m/s}^2 = 1\text{ N}]$$

57. i. The reason is that one cannot exert an action force effect on the slippery peel of banana in the backward direction. Hence, in response the ground does not exert sufficient reaction force in forward direction and hence we lose our balance.
- ii. This can be explained on the basis of the law of conservation of momentum. When due to the explosion some pieces move in the same particular direction, then in order to conserve momentum the remaining pieces move in the opposite direction.
- iii. The reason is that the glass pane of the window is a hard solid. The flying pebble suffers a change in momentum in a very short time, so the force exerted by the glass window on the pebble will be large. Consequently, the glass pane of the window will shatter.
- iv. The reason is that the mass of the cricket ball is more than that of a tennis ball. Thus, momentum is more in the case of the cricket ball due to the larger mass as compared to the tennis ball. So, less force has to be applied in the case of the tennis ball to stop.
- v. The reason is that when athletes are running for the throw, then due to the inertia of motion they often fail to stop themselves before the line.

58. Consider an object of mass 'm' moving along a straight line with an initial velocity 'u' (say). It is uniformly accelerated to velocity 'v' in time t by the application of a constant force 'F' for time 't'.

Then, initial momentum of the object = mu

$$\therefore p_1 = mu$$

Final momentum of the object = mv

$$\therefore p_2 = mv$$

$$\therefore \text{Change in momentum} = p_2 - p_1 = mv - mu = m(v - u)$$

$$\text{The rate of change in momentum} = \frac{m \times (v - u)}{t}$$

According to Newton's second law of motion, we have

$$F \propto \frac{M(v-u)}{t}$$

$$F = kM \frac{(v-u)}{t}$$

$$\text{Since, } a = \frac{v-u}{t}$$

$$\text{Therefore, } F = k \times m \times a \text{ -----(i)}$$

Here,

a = acceleration

k = a constant of proportionality.

When m = 1 kg, a = 1 ms⁻² then F = 1N

$$1N = 1Kgm s^{-2}$$

Let us substitute these values in equation (i), we get

$$1N = k \times 1kg \times 1ms^{-2}$$

$$\therefore K = 1$$

From equation (1), we have

$$F = ma$$

This represents the second law of motion.

Thus, the second law of motion gives a method to measure the force acting on an object as a product of its mass and acceleration.

59. **Balanced forces:** If the resultant of applied forces is equal to zero, it is called balanced forces.

Example: In the tug of war if both the teams apply similar magnitude of forces in opposite directions, the rope does not move on either side. This happens because of balanced forces in which resultant of applied forces become zero.

Unbalanced force: If the resultant of applied forces is greater than zero the forces are called unbalanced forces.

An object in rest can be moved because of applying balanced forces. According to Newton's first law of motion, the acceleration of the body could be zero if there is no force acting on it. Acceleration in an object could be zero even when several forces acting on it. This happens when all the forces cancel out each other i.e. the net force acting on the object is zero. For a uniformly moving object, the net force acting on the object is zero. Hence, the acceleration of the object is zero.

60. **Newton's Second Law of Motion:** The rate of change of momentum is directly proportional to the force applied in the direction of force.

According to the question:

Initial velocity (u) = 0, final velocity (v) = 10m/s, time (t) = 20 second, Mass (m) = 10000 kg,

Therefore, force (F) = ?

$$\text{We know that force (F)} = m \frac{(v-u)}{t}$$

$$F = (10000) \frac{(10-0)}{20}$$

$$= \frac{100000}{20}$$

$$= 5000 \text{ N}$$

Thus the required force is 5000 N.

61. According to law of conservation of momentum; the momentum of bullet forward will be equal to the momentum of rifle backward. In case of light rifle; velocity will be more than the velocity of heavier rifle because of mass so that momentum (product of mass and velocity) for both shall not be equal. Due to this, the lighter rifle will hurt the shoulder more. In both cases it has been called recoiled.

62. Momentum can be defined as "mass in motion". All objects have mass, so if an object is moving, then it has momentum, which means it has its mass in motion. It is the product of the mass and velocity of the body.

It is denoted by p.

Momentum, $p = mv$

Where

p = momentum, m = mass, v = velocity

The equation illustrates that momentum is directly proportional to the mass of an object and directly proportional to the velocity of that object.

Its SI unit is kg-m/s.

Force applied on an object of mass m moving with acceleration a,

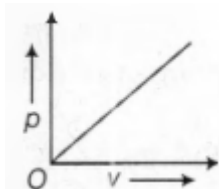
$$F = ma = m \frac{\Delta v}{\Delta t} \quad [\because \text{acceleration} = \text{rate of change of velocity} = \frac{\Delta v}{\Delta t}]$$

$$= \frac{\Delta p}{\Delta t}$$

$\therefore$ Force applied on an object is equal to the rate of change of momentum of the object.

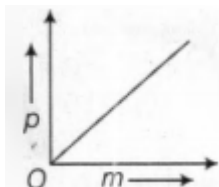
i. Momentum versus velocity graph when mass is fixed.

If m is fixed, then $p \propto v$



$\therefore$ Momentum versus velocity graph will be a straight line passing through the origin (if $v = 0$, then $p = 0$).

ii. Momentum versus mass graph when velocity is constant.



If velocity is constant, then $p \propto m$.

So, the momentum versus mass graph will be a straight line passing through the origin (if $m = 0$, then $p = 0$).

63. Mass of player 1 = $m_1 = 60 \text{ kg}$

Mass of player 2 = $m_2 = 55 \text{ kg}$

If v is the velocity of the two entangled players after the collision, the total momentum then

$$= (m_1 + m_2) \times v$$

$$= (60 + 55) \text{ kg} \times v \text{ ms}^{-1}$$

$$= 115 \times v \text{ kg ms}^{-1}$$

Equating the momenta of the system before and after the collision, in accordance with the law of conservation of momentum, we get

$$v = \frac{-30}{115}$$

$$= -0.26 \text{ ms}^{-1}$$

Thus, the two entangled players would move with velocity 0.26 m s^{-1} from right to left, that is, in the direction the second player was moving before the collision.

Let the first player be moving from left to right. By convention left to right is taken as a positive direction and thus right to left is the negative direction. If symbols m and u represent the mass and initial velocity of the two players, respectively. Subscripts 1 and 2 in these physical quantities refer to the two hockey players. Thus,

$$m_1 = 60 \text{ kg}; u_1 = + 5 \text{ ms}^{-1}; \text{ and}$$

$$m_2 = 55 \text{ kg}; u_2 = - 6 \text{ ms}^{-1}.$$

The total momentum of the two players before the collision

$$= 60 \text{ kg} \times (+ 5 \text{ m s}^{-1}) + 55 \text{ kg} \times (-6 \text{ ms}^{-1})$$

$$= - 30 \text{ kg ms}^{-1}$$

64. According to Newton's first law of motion, a body at rest or in uniform motion will remain at rest or in uniform motion unless an unbalanced force acts upon it. Newton's first law defines inertia and is rightly called the law of inertia. This law consists of three parts:

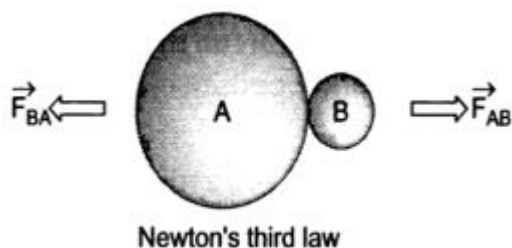
(i) The first part says that a body at rest continues in its state of rest. For instance, a boy standing in a train falls backward when the train suddenly starts moving forward. This is because when the bus moves, the lower part of his body begins to move along with the train while the upper part of his body continues to remain at rest due to inertia.

(ii) The second part says that a body in uniform motion continues to move in straight line path with a uniform speed, e.g., when a moving train stops suddenly a person sitting in it falls forward. This is because as the train stops, the lower part of the person's body comes to rest along with the bus while upper part of his body continues to remain in motion due to inertia of motion and thus he falls forward.

(iii) Third part says that a body moving with a uniform speed in a straight line cannot change its direction of motion by itself. For example, when a bus takes a sharp turn, a person sitting in the bus gets force acting away from the centre of the curved path due to his tendency to move in the original direction.

65. According to Newton's third law of motion, 'To every action, there is an equal and opposite reaction; action and reaction forces act on different bodies.'

When an object, say A, exert a force (action) on another object, say B, then B also exerts a force (reaction) on the A. These two forces are always equal in magnitude but opposite in direction.



As shown in the above figure, if F_{AB} be the force exerted by body A on B and F_{BA} is the force exerted by B on A, then according to Newton's third law of motion,

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

or, Force on B by A = Force on A by B

or, Action = Reaction

This law clarifies that a single force can never exist and that the forces always exist in pairs. The two opposing forces are known as action and reaction. The forces of action and reaction always act on two different bodies.

66. Force, $F = 5 \text{ N}$

Mass, $m = 2 \text{ kg}$

i. $F = m \times a$

$$\Rightarrow 5 = 2 \times a$$

$$\Rightarrow a = 2.5 \text{ m/s}^2$$

Therefore, acceleration produced by the body is 2.5 ms^{-2}

- ii. Final velocity, $v = 0$

Initial velocity, $u = 0$ (body starts from Rest)

Time, $t = 4 \text{ s}$

From the relation,

$$v = u + at$$

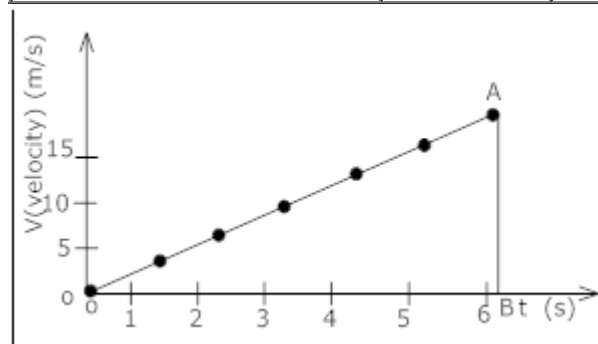
$$\Rightarrow v = 2.5 \times 4$$

$$\Rightarrow v = 10 \text{ m/s}$$

Therefore, the velocity at $t = 4 \text{ s}$ is 10 ms^{-1}

iii.	For $t = 0$	1	2	3	4	5	6
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V = 0	2.5	5	7.5	10	12.5	15
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iv. Distance Travelled = Area under v/t curve = Area of $\triangle AOB$

$$= \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$= \frac{1}{2} \times OB \times AB$$

$$= \frac{1}{2} \times 6 \times 15$$

$$= 45 \text{ m}$$

Therefore, distance travelled in 6 s is 45 m.

67. **Newton's First Law of Motion:** Any object remains in the state of rest or in uniform motion along a straight line until it is compelled to change the state by applying an external force.

Explanation: If an object is in the state of rest, then it will remain in rest until an external force is applied to change its state.

Similarly, an object will remain in motion until an external force is applied over it to change its state. This means all objects resist changing their state. The state of any object can be changed by applying external forces only.

A person standing on a bus falls backward when the bus starts moving suddenly. This happens because the person and bus both are in rest while the bus is not moving, but as the bus starts moving the legs of the person start moving along with the bus but the rest portion of his body has a tendency to remain in rest. Because of this person falls backward; if he is not alert.

Newton's Second Law of Motion: The rate of change of momentum is directly proportional to the force applied in the direction of the force.

Explanation: Newton's Second Law of Motion gives the relation between force, mass and acceleration of an object.

For example, when acceleration is applied on a moving vehicle, the momentum of the vehicle increases and the increase is in the direction of motion because the force is being applied in the direction of motion. On the other hand, when the brake is applied on the moving vehicle, the momentum of the vehicle decreases and the decrease is in the opposite direction of motion because the force is being applied in the opposite direction of motion.

Newton's Third Law of Motion: There is an equal and opposite reaction for every action

Explanation: Whenever a force is applied over a body, that body also applies the same force of equal magnitude and in the opposite direction.

Example: Walking of a person: A person is able to walk because of Newton's Third Law of Motion. During walking, a person pushes the ground in the backward direction and in the reaction the ground also pushes the person with an equal magnitude of force but in the opposite direction. This enables him to move in the forward direction against the push.

68. **Force:** Push or pull of an object is considered a force.

Example: to open a door, either we push or pull it. A drawer is pulled to open and pushed to close.

Effect of Force:

- Force can make a stationary body in motion. For example, a football can be set to move by kicking it, i.e. by applying a force.
- Force can stop a moving body: For example, by applying brakes, a running cycle or a running vehicle can be stopped.
- Force can change the direction of a moving object. For example; By applying force, i.e. by moving handle the direction of a running bicycle can be changed. Similarly by moving steering the direction of a running vehicle is changed.
- Force can change the speed of a moving body: By accelerating, the speed of a running vehicle can be increased, or by applying brakes the speed of a running vehicle can be decreased.
- Force can change the shape and size of an object. For example: By hammering, a block of metal can be turned into a thin sheet. By hammering a stone can be broken into pieces.

69. Force exerted by the engine, $F' = 40,000 \text{ N}$

Frictional force offered by the track in the direction opposite of the motion, $F'' = -5,000 \text{ N}$

(a) The net accelerating force, $F = F' + F'' = 40,000 \text{ N} + (-5,000 \text{ N}) = 35,000 \text{ N}$

(b) Mass of each wagon of the train = 2000 kg

Number of wagons = 5

Therefore, Mass of the train, $m = 2,000 \text{ kg} \times 5 = 10,000 \text{ kg}$.

Net accelerating force acting on the train, $F = 35000 \text{ N}$

From Newton's second law of motion, acceleration

$$a = \frac{F}{m} = \frac{35,000 \text{ N}}{10,000 \text{ kg}} = 3.5 \text{ ms}^{-2}$$

(c) Mass of 1 wagon = 2000 kg

acceleration of the train = 3.5 ms^{-2}

From the relation, $F = ma$, we get

$$F = 2000 \text{ kg} \times 3.5 \text{ ms}^{-2}$$

$$F = 7000 \text{ N}$$

Force exerted by wagon 1 on wagon 2

$$= \text{Net accelerating force} - \text{Force acting on wagon 1} = 35000 \text{ N} - 7000 \text{ N} = 28000 \text{ N}$$

Therefore, the required answer is 28000 N

70. i. In the beginning water and clothes are in the state of rest. When the wet clothes are jerked, the particles of water tend to continue in its state of rest because of inertia of rest. Therefore the removal of water helps wet clothes to dry quickly because the droplets of water fall off from the clothes.
- ii. When a carpet is hit with a stick, the carpet moves, but dust on account of inertia of rest flies off because of the sudden movement. It is in fact the carpet which comes out of the dust.
- iii. In the beginning the fruits on the branches are in the state of rest. On account of inertia of rest, the fruits tend to continue in their state of rest, but due to strong wind the fruits fall off the branches.
- iv. At first the pillion rider and driver are in a state of motion. When the driver applies brakes, the body of pillion rider continues moving forward on account of inertia of motion. Hence the pillion rider falls forward.